
Course Overview

Modules

- Data and Systems – 4 weeks
- Pixel Processing – 1.5 weeks
- Color Management – 2 weeks
- Morphological Operators -1.5 weeks
- Image Compression – 0.5 weeks

Use of AI/ML

- This is fundamental DIP
 - Fundamentals don't need AI/ML
- Examples of how AI/ML can be used to improve performance
- AI/ML is not a prerequisite for this course

Assignments and Exams

- Each module has WA (Easy)
 - WA is graded for completeness and not correctness
 - Solutions will be available as soon as you submit
 - 5% of your grade
- Each module has PA
 - 40% of grade
- Beginning of the week pop quiz (Easy)
 - Absolutely fundamental concepts
 - 5% of grade

Assignments and Exams

- One Midterm
 - 20% of grade
- One Final
 - 25% of grade
- Attendance is mandatory (Easy)
 - 5% of your grade
- 15% of your grade is for you to be engaged
- 85% of your grade is for your hard work

Performance and Learning

- Grades are only one metric
- Have you learned the course? Can you ace interviews?
 - Extra credits in written assignments
- No one is naturally talented to ace this course

Module 1: Data and Systems

- Data and LSIS
- Convolution and Filters
- Feature Detectors
- Spectral Analysis – 1D
- Spectral Analysis – 2D (Image)

Module 2: Pixel Processing

- Geometric Transformations
 - Linear and Non-Linear Transformations
 - Estimating Transformations
- Statistical Color Processing
 - Histogram Processing
- Image Composition

Module 3: Color Management

- Device Independent Color Space
- Color Mixtures
- Color Management

Module 4: Morphological Operators

- Binary Images
 - Erosion and Dialation
 - Opening and Closing
- Gray Scale Images
- Applications

Module 5: Image Compression

- Different Compression Techniques
- Evaluating Compression
- JPEG Compression
- Basics of MPEG Compressions

Schedule and People

- Instructor – Aditi Majumder (majumder@ics.uci.edu)
- Class: MW 9:30am-10:50pm – DBH 1300
- TA – Jiabao Li (jiabaol6@uci.edu)
- Instructor Office Hours (Calit2, Rm. 2301)
 - Friday 3-4pm
- TA Office Hours (DBH, Rm. 4211)
 - Tuesday – 10am - 11am
 - Thursday – 2pm - 3pm
- You have contact hours *every day of the week*
- All PA questions should be brought to the TA

Exams

- Midterm 1: April 22 (In class)
- Midterm 2: May 18 (In class)
- Finals : June 10, Wed, 8am-10am, DBH 1300

We won't administer absentee exam